

ISO 9001:2015 Quality Management System Auditor

- A few words about ISO 9001-



ISO 9001 – first publication in 1987
4 revisions – 1994, 2000, 2008, 2015

ISO 9001 is applicable to any type and size of organization
&
Over 1.000.000 organization have implemented QMS acc. to ISO 9001

QMS – A set of of processes, policies and procedures needed by
an organization to plan and perform its activities (regardless
whether its production or service provision)

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- The ISO 9000 family -



ISO 9000:2015 - Fundamentals and vocabulary

ISO 9004:2009 - Managing for the sustained success of an organization -- A quality management approach

ISO 19011:2011 - Guidelines for auditing management systems

ISO/TS 9002:2016 - Guidelines for the application of ISO 9001:2015

- ISO 10001:2007 - Customer satisfaction -- Guidelines for codes of conduct for organizations
- ISO 10002:2014 - Customer satisfaction -- Guidelines for complaints handling in organizations
- ISO 10003:2007 - Customer satisfaction -- Guidelines for dispute resolution external to organizations
- ISO 10004:2012 - Customer satisfaction -- Guidelines for monitoring and measuring
- ISO 10005:2005 - Quality management systems -- Guidelines for quality plans
- ISO 10006:2003 - Quality management systems -- Guidelines for quality management in projects

www.iso.org

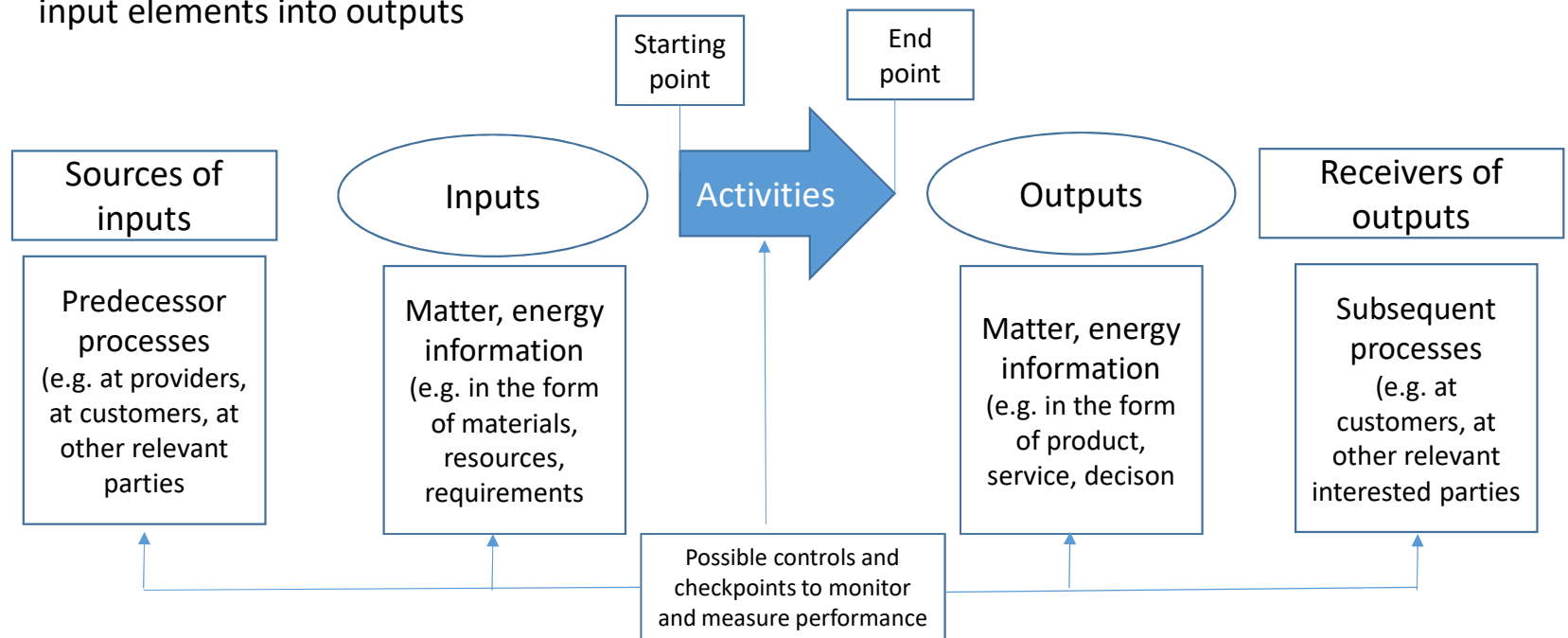
ISO 9001 Audit Practices Group - an informal group of quality management system experts, auditors and practitioners

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- Process approach -



Process = a set of activities which are usually interrelated in interconnected that transform input elements into outputs



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- Process approach -



Possible classification of processes:

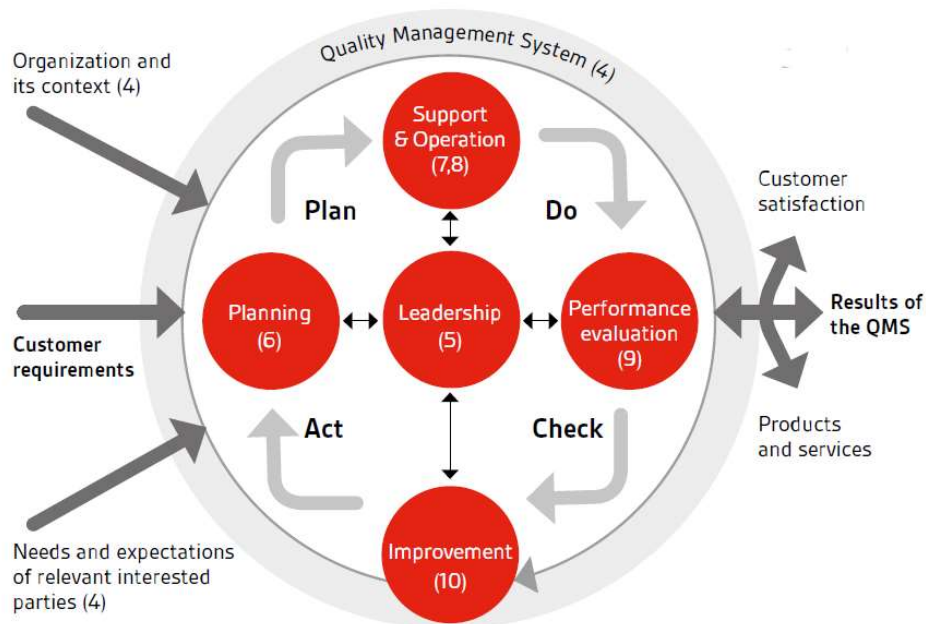
- Management processes – related to the top management (provision of resources, defining the policy of the organization, defining roles and responsibilities for staff)
- Product/ service realization processes – they add value to the organization (main processes)
- Supporting processes – sustain the product/ service realization processes
- Measuring and monitoring processes

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- Process approach -



PDCA CYCLE PLAN – DO – CHECK - ACT



Plan: define objectives, the resources needed, identify risks and opportunities for the process

Do: implement what was planned

Check: monitor and measure the process and its outputs – do they respect the requirements set in the planning phase

Act: take actions to improve performance, as necessary

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- Risk based thinking -



Risk = effect of uncertainty



Risk based thinking – part of every process – it cannot be implemented or audited as a stand-alone requirement

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- Understanding the organization and its context -



External issues: legal requirements, other requirements;
technological aspects; market related aspects; social aspects;
economic and cultural aspects

Internal issues: values and culture of the organization; HR;
knowledge; equipment and technology

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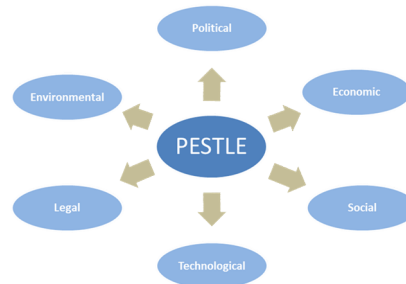
- Understanding the organization and its context -



AUDITING

- Interviews with top management
- Documented information: SWOT analysis; PESTLE analysis; market analysis; marketing plans, etc

External and internal issues should be continuously reviewed as they are in constant change



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- Understanding the needs and expectations of interested parties -



- IDENTIFY INTERESTED PARTIES
- DETERMINE THE REQUIREMENTS OF THE INTERESTED PARTIES THAT ARE RELEVANT TO THE QMS

Examples of interested parties: customers, end users of products and services, partners, franchisors, owners, shareholders, bankers, unions, suppliers, employees or persons working on behalf of the organization, authorities, NGOs, community, competitors, trade associations.

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- Understanding the needs and expectations of interested parties -



Requirements of interested parties may be found or identified :

- orders and contracts,
- industry codes and standards,
- conventions and protocols,
- communication with the community,
- legal and regulatory requirements applicable to the activity, to products and services;
- participation in associations;
- benchmarking and market surveillance,
- reviewing the relations with suppliers,
- conducting customer reviews,
- monitoring end-user satisfaction reports

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- Understanding the needs and expectations of interested parties -



AUDITING

No requirement for specific documented information

Auditor may conduct interviews and may review any documents available

+

Management review

Needs and expectations of interested parties do change in time.

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- Determining the scope of the QMS -



QMS Scope = what processes and locations from the organization are included in the Quality management system

Elements to take into consideration when determining the QMS Scope:

- The external and internal issues
- The requirements of the interested parties
- The products and services of the organization

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- Determining the scope of the QMS -



Some requirements of ISO 9001:2015 may be considered not applicable

- Those requirements considered not applicable shall not affect the organization's ability to provide products and services that meet requirements or enhance customer satisfaction
- There has to be a justification for considered some requirements not applicable

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- QMS scope should be maintained as documented information (may be included in a document like a quality manual or may be a stand alone document)
- Look at the justification of requirements considered not applicable

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- Quality management system and its processes -



Process = set of activities to transform inputs into outputs

For each process the organization has to:

- Determine INPUTS and OUTPUTS.
- Determine the SEQUENCE and INTERACTION of processes.
- Determine and apply CRITERIA TO CONTROL the processes.
- Determine RESOURCES for each process.
- Assign RESPONSIBILITIES and AUTHORITIES for each process.
- Address the RISKS and OPPORTUNITIES for the process.
- EVALUATE the processes and ensure that if changes are needed to make the process achieve intended results.
- IMPROVE the processes.
- MAINTAIN documented information to sustain operation.
- RETAIN documented information to ensure processes are carried out as planned.

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- Quality management system and its processes -



MAINTAIN vs. RETAIN

MAINTAIN –documented information that may be subject to revisions – like procedures, manuals, guidelines, software, apps, specifications, work instructions.

RETAIN - documented information normally retained unchanged to demonstrate conformity and that the processes are carried out as planned.

AUDITING

Identification of QMS processes and their succession & interaction
– may be drawing of schematic representation.

Inputs, outputs, authorities, responsibilities, risks, opportunities, resources, criteria for control, documented information.

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- Leadership and commitment -



Top management has to be actively involved in the implementation and improvement of the QMS

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- Leadership and commitment -



Top management:

- Takes accountability for the QMS
- Ensures quality policy & objectives
- Ensures integration of QMS in business processes
- Promotes process approach & risk based thinking
- Provides resources for the QMS
- Communicates importance of quality and conformity
- Ensures QMS achieve intended results
- Engages and supports others for the effectiveness of QMS
- Promotes improvement
- Supports others to demonstrate leadership

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- Leadership and commitment -



AUDITING

Top management involvement is vital for the QMS

Interviews with top management & during the audit with organization personnel

Review of documented information – quality policy, objectives, mission, values, budgeting including resources for the QMS

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- Customer focus -



It is also top management responsibility to ensure that:

- Customer, legal & other requirements are – determined, understood & met
- Risks and opportunities that affect conformity are addressed
- Maintain focus on enhancing customer satisfaction

Customers: clients, end-users, patients, students, citizens, etc

AUDITING

Interviews with top management

Review of documented information – customer satisfaction monitoring, controls of authorities, complaints and legal actions, public information of the company on the internet/ newspapers, etc

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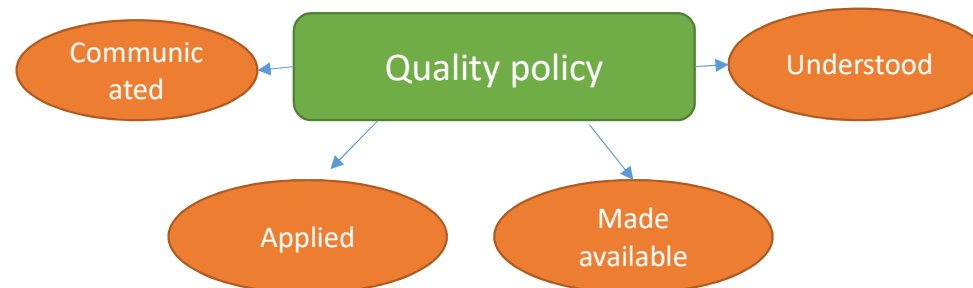
- Establishing and communicating the quality policy -



Quality policy - the intensions and direction of the organization

Quality policy:

- Appropriate to the context of the organization
- Provides a framework for quality objectives
- Includes commitment to satisfy applicable requirements
- Includes commitment for continual improvement



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- Establishing and communicating the quality policy -



AUDITING

QP should be available as documented information and include required elements (commitments)

Look for how it is communicated in the organization – trainings, posters

Interviews with personnel to evaluate whether they are aware of the existence of the QP and understand it

Look for how it is made available to interested parties – website, brochures, provided upon request...

Top management should review QP periodically to ensure it stays relevant – documented information of management review or other meetings to discuss the QP

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- Organizational roles, responsibilities and authorities -



Top management – establishes and communicates roles, responsibilities, authorities

Responsibilities and authorities for the QMS:

- To ensure that the QMS conforms to the requirements of ISO 9001
- To ensure that processes are delivering intended outputs
- To report to top management on performance of the QMS and opportunities for improvement
- To ensure the promotion of customer focus in the organization
- To ensure that the QMS integrity when changes are implemented

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- Organizational roles, responsibilities and authorities -



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System that is used to establish and communicate roles, responsibilities and authorities – job descriptions, work instructions, duty statements, internal regulations, organizational charts, decisions

Identity of person(s) who have responsibility and authority for the QMS :

- What are their specific responsibilities and authorities?
- Do they understand the responsibilities and authorities they have?

Review of documented information + interviews

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- Actions to address risks and opportunities -



Requirement

Determine risks and opportunities and plan actions to address them

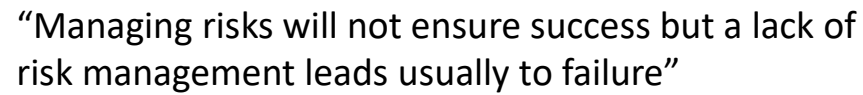
Why?

To prevent nonconforming outputs & find opportunities to enhance customer satisfaction or achieve quality objectives

What to take into consideration?

CONTEXT + INTERESTED PARTIES

RiG
CERT
TRAINING



Risk management - understanding the uncertainty elements and to act in order to affect this uncertainty

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- Actions to address risks and opportunities -



Techniques:

SWOT analysis (Strength, Weaknesses, Opportunities and Threats),
PESTLE (Political, Economical, Social, Technological, Legal, Environmental)
FMEA (Failure Mode and Effects Analysis)
HACCP (Hazard Analysis and Critical Control Points).
Matrix of consequences and probabilities
What if techniques.
Brainstorming.

ISO 31000 family – risk management
IEC 31010 – risk assessment

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- Actions to address risks and opportunities -



RISKS AND OPPORTUNITIES ARE SPECIFIC TO EVERY ORGANIZATION

Risks:

HR related risks

Market related risks

Infrastructure related risks

Financial risks

Legal compliance related risks

Opportunities:

adopting new technologies,

going on new markets,

launching new products,

establishing partnerships

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- Actions to address risks and opportunities -



Possible actions:

Avoid the risk

Eliminate the risk

Take the risk

Share the risk

Take no action

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- Actions to address risks and opportunities -



AUDITING

Clarify if there is a formal documented risk management process
(possible for other reasons – ex. legal)
Then documented information is available (ex. risk assessment)

In case formal documented information on risk management is
not maintained:

- Interviews with top management and personnel
- Use of other documented information – management programs, management reviews, strategy plans, etc

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- Quality objectives and planning to achieve them-



Quality objectives = maintained as documented information

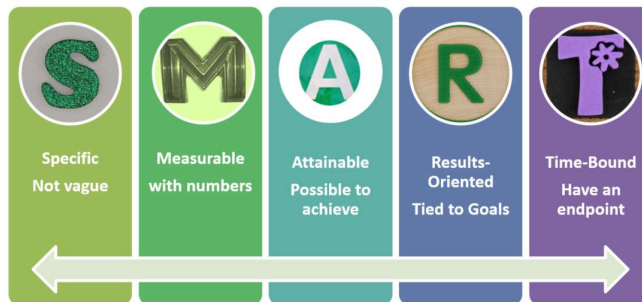
Set at relevant functions, levels and processes

Quality objectives have to:

- Be consistent with the quality policy.
- Be measurable.
- Take into account applicable requirements.
- Be relevant to the conformity of products and services and to the enhancement of customer satisfaction.
- Be monitored.
- Be communicated.
- Be updated as necessarily.

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- Quality objectives and planning to achieve them-



When defining quality objectives the organization needs to take into account its current capabilities, customer feedback, market issues its constraints.

Actions to achieve objectives:

What will be done

What resources will be required and make them available

Who is responsible

When it will be completed;

How the results will be evaluated.

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- Quality objectives and planning to achieve them-



Example of a door selling company expanding its range of products with fireproof doors

Objective	Actions	Resources	Responsibilities	Time to complete	How results are evaluated
Expand the range of doors with fireproof doors	Market research and identification of 3-5 suppliers	2 days	John from marketing	Until 15.01.2017	List of potential suppliers
	Negotiation and contract with 1-3 suppliers	1-5 days	Purchasing manager	25.02.2017	Contracts signed
	Aquisition of 5 different fireproof doors for showroom	3000 Euros	Purchasing manager	28.02.2017	Purchased doors present in the showroom
	Updating the online and paper catalogue by including those products	1000 Euros	John from marketing	10.03.2017	Paper catalogue and online catalogue
	Online marketing on fireproof doors	500 Euros	Online marketing agency in cooperation with John	Until 01.05.2017	Ads, campaigns

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- Quality objectives and planning to achieve them-



AUDITING

Quality objectives – documented information

- Responsibilities
- Resources available
- Specific actions and timeframes

Documented information – may be stand-alone document or strategies, plans, project management, management review meetings, etc

Interview with top management

Interview with personnel to see whether they are aware of the quality objectives and their individual tasks in achieving objectives

Accomplishment is monitored?

Objectives are updated?

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- Planning of changes -



Change can have
positive or negative effects

Examples: moving from a site to another, changing processes or methods used, changing the information technology or the software used, outsourcing, key personnel leaving, switching to online environment, opening a new office in a different location, etc

Proper planning of change – helps
avoiding negative consequences

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- Planning of changes -



When planning to change the organization should use Risk Based Thinking

- Evaluate the impact of change
- Take actions to prevent undesired effects

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Is there a formal documented change management process?
(documented information)

Are responsibilities assigned for coordinating change?

Other documented information: plans, strategies, project management, minutes, risk evaluations.

Interview with management and people involved in the change process.

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- Resources -



Determine and provide the resources needed for the establishment, implementation, maintenance and continual improvement of the QMS

RESOURCES

People



Infrastructure



Environment for processes



Knowledge



Measuring & monitoring resources



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- People -



Organization has to determine and provide the needed human resources for the operation and control of its processes and for the QMS

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- Human resources are appropriate to the needs (in terms of number and competence)?

(interviews, observation, review of documents)

- Turnover rate – is it identified as risk?
- Subcontracting and leasing of personnel – their training and awareness

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Determine, provide and maintain the infrastructure needed to ensure operation of its processes and to achieve conformity of products and services.

Infrastructure:



Buildings & utilities

Equipment – hardware and software

Transportation resources

Information and communication technology

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AUDITING

No specific requirement for documented information.

How is it maintained? Using own forces or subcontractors, or both?

Is there preventive maintenance or only corrective?

Personnel performing maintenance is competent? (authorizations, licenses)

Documented information of the infrastructure maintenance process

Interviews with personnel in charge with maintenance

Risks and opportunities related to infrastructure?

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- Environment for the operation of processes -



Organization is required to determine, provide and maintain the right environment needed to operate its processes and to achieve conformity of products and services

Human factors

Physical factors

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No requirement for documented information.

Specific environment requirements for the sector.

Health & Safety matters and medical testing of employees.

Monitoring of physical factors – temperature, humidity, etc

Actions taken to reduce stress, burnout, to provide a calm environment.

Observation + interview

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- Monitoring and measuring resources -



The organization needs to determine and provide the resources to monitor and measure the conformity of products and services.

They need to be:

- Suitable
- Maintained properly

Documented information on the fitness for purpose of such resources has to be retained.

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- Monitoring and measuring resources -



Equipment for measuring is used when measurement traceability is required or considered essential.



Equipment has to be:

- Calibrated or verified , or both
- Identified
- Protected

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- Monitoring and measuring resources -



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What resources are used and review documented information for their fitness for purpose.

Equipment is used to measure?

If so is it calibrated and/ or verified or both?

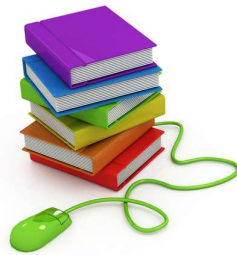
Is the status of calibration/ verification identified?

Is the equipment protected against actions that may affect its calibration?

In case the organization did subcontract the measurement and monitoring activity then how does the organization monitor the competence and performance of the external provider.

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- Organizational knowledge -



The organization is required to determine, maintain and make available the knowledge necessary for the operation of its processes and to achieve conformity of its products and services.

Sources:

- Learning from successes or failures, or near miss situations or incidents.
- Knowledge obtained from partners, clients, suppliers, consultants.
- Benchmarking.
- Documented knowledge of people in the organization, or making it available through training or mentoring.
- Libraries, access to websites, access to online storage, subscriptions, etc.

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- Organizational knowledge -



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What system is used by the organization to gather and manage knowledge and how is it available when needed.

The options belong to the organization.

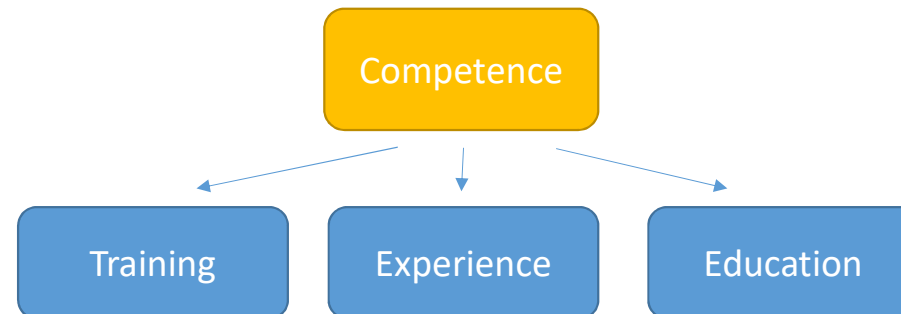
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- Competence -



The organization is required to:

- Determine the competence needed
- Ensure personnel is competent
- Take actions to acquire needed competence and evaluate the effectiveness of those actions
- Retain documented information as evidence of competence



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AUDITING

Review of documented information

*(diplomas, authorizations, permits, certificates, licenses, resumes;
training evidence, etc)*

Evaluate the training process (internal & external) and the methods to
verify the effectiveness of training

Competence of personnel belonging to subcontractors & external
providers

Competence is a highly valuable asset – risks and opportunities
related to competence?

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- Awareness -



People under the control of the organization need to be aware of:

- the quality policy
- quality objectives
- their contribution to the effectiveness of the quality management system, including the benefits of improved performance;
- the implications of not conforming with the quality management system requirements.



Communication for awareness

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- Awareness -



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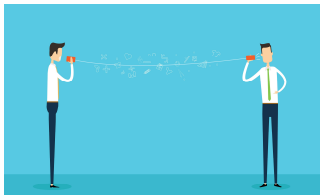
Review of documented information: training records, written communication

Interviews to evaluate level of awareness

Observation of actions and behavior

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- Communication -



Determine external and internal communications.

On what it communicates?

When it communicates?

With whom it communicates?

How it communicates?

Who communicates?

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AUDITING

Any procedure for communication that describes the methods?

Communication methods are fit for purpose?

Interview to evaluate internal communication

Documented information to support the communication process

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- Documented information -



The QMS documentation has to include:

- Documented information requested by ISO 9001 (whenever the standard says maintain or retain documented information).
- Documented information determined by the organization as being necessary for the effectiveness of the quality management system.

Size of documented information depends on:

- size and complexity of operations and processes,
- legal and regulatory requirements applicable and
- competence of personnel

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- Documented information -



Creating & updating:

- Identification and description
- Format
- Review and approval.

Control of documented information to ensure:

- it is available in a suitable format whenever needed;
- it is adequately protected.

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- Documented information -



Control of documented information:

- distribution, access, retrieval and use
- storage and preservation
- control of changes
- retention and disposition

External origin documents may include – documented information from customers, external providers, legal and regulatory requirements, standards, guides – any documented information that has not been created by the organization and that is needed for the planning and operation of the QMS. It should be identified and protected similar to the documented information elaborated internally.

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- Documented information -



AUDITING

Documented information required to be maintained specifically by the standard: scope, quality policy, quality objectives

+

Documented information it has considered necessary to maintain to ensure the effectiveness of its management system (like manuals, procedures, instructions, etc)

Responsibilities for reviewing and approving documented information have been assigned?

How documented information is controlled: storage, protection, access, retrieval, use, distribution, control of changes, retention periods, disposing of obsolete documents.

Usually this requirement is not evaluated as stand-alone. Documented information maintained or retained is evaluated while other requirements are being audited.

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- Operational planning and control -



The organization is required to:

<u>Plan</u>	the processes necessary
<u>Implement and</u>	for the provision of its
<u>Control</u>	products and services.

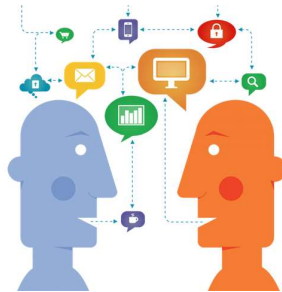
It has to:

- Determine the requirements for its products and services.
- Establish criteria for its processes and for the acceptance of products and services.
- Determine what resources are needed to achieve conformity of its products and services.
- Implement controls according to the criteria for products and services established.
- Maintain and retain documented information to have confidence that processes have been carried out as planned and to demonstrate the conformity of its products and services.

+ Control changes and outsourced processes

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- Customer communication -



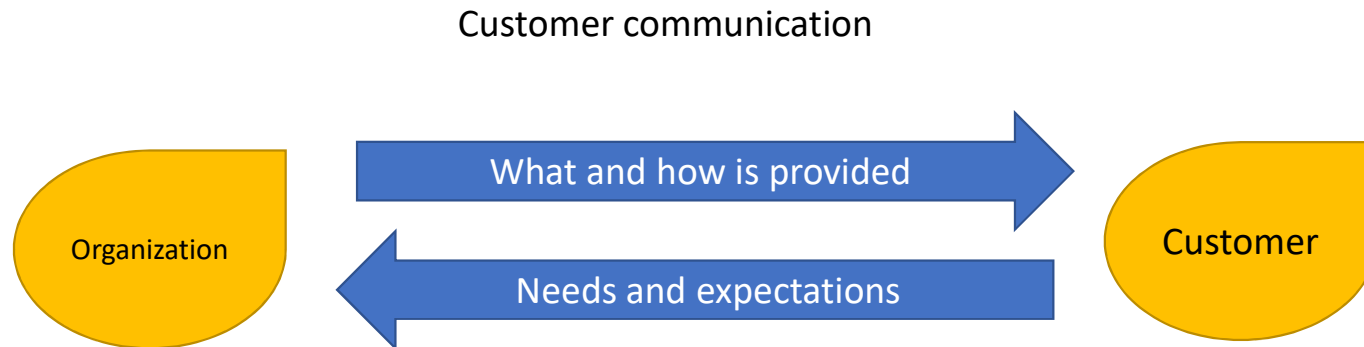
Communication between the organization and its customers has to be clear and efficient

The organization is required to:

- Provide information.
- Handle enquiries, contracts and orders including changes.
- Obtain feedback.
- Inform on customer property aspects.
- Establish specific communication for contingency actions.

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- Customer communication -



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- Customer communication -



AUDITING

Responsibilities for customer communication have been assigned?

Accuracy and completeness of information about products and services – websites, brochures, product specification advertising.

Procedures to respond to communications from the customer – *the complaint handling process has to be evaluated (review, responsibilities, timeframes).*

Effectiveness of communication – language, cultural aspects.

Review of documented information and interview

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- Determining requirements for products and services -



The organization shall have a clear image of the requirements applicable to the products and services that it intends to place on the market.

Aspects to be taken into consideration:

- purpose of its products and services
- needs and expectations of its intended customers
- legal and regulatory requirements apply to its products and services
- requirements considered necessary by the organization.

The organization has to be able to meet the claims it makes for its products and services.

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- Determining requirements for products and services -



AUDITING

The process to determine requirements for its products and services

The claims on products and services and how those claims are being covered (sample).

Interviews can be used here and review of documented information like (brochures, offers, websites, etc)

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- Review of the requirements for products and services -



The organization is required to:

- ensure it is able to meet the requirements for products and services being offered
- conduct a review before committing to supply products and services

The review should include:

- Requirements of the client
- Requirements not explicitly stated by the client but necessary for the intended use
- Additional requirements of the organization
- Statutory and regulatory requirements
- Changes made to initial orders/ contract

Retain documented information on:

- Results of review
- Any new requirements for products & services

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- Review of the requirements for products and services -



Changes to customer requirements

In case of changes to requirements:

- Documented information is amended
- Relevant persons are made aware.

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The system used for reviewing customer requirements prior to committing to provide.

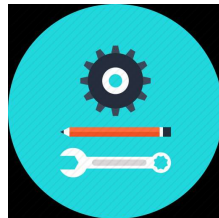
Responsibilities and authorities for conducting the review and making decisions.

The management of changes to orders/ contracts – how they are handled by the organization

Review documented information.

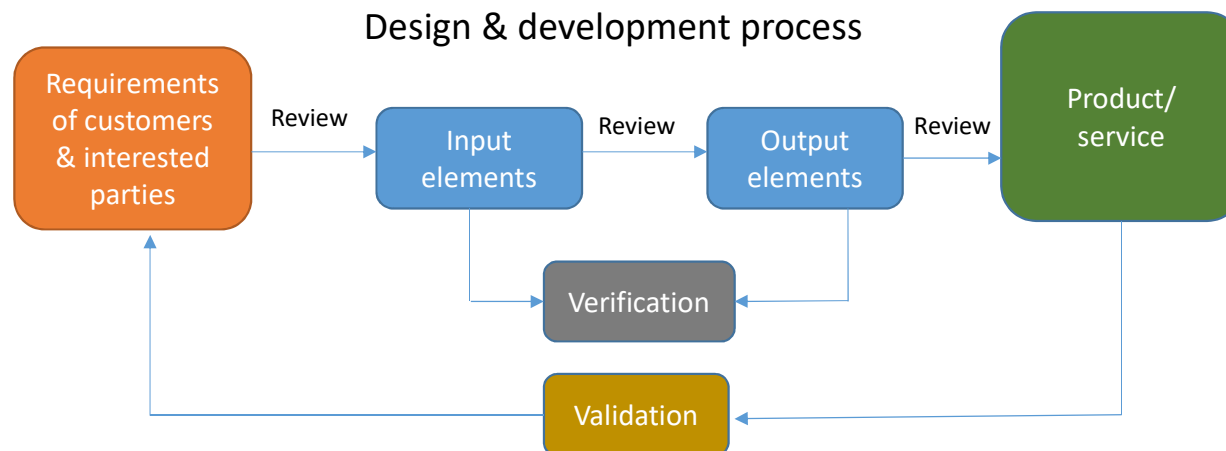
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- Design and development of products and services -



The organization shall establish, implement and maintain a design and development process that is appropriate to ensure the subsequent provision of products and services.

Applicable / Not applicable ?



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- Design and development planning -



The organization has to plan the design and development process in order to determine the stages of this process as well as the controls needed.

The following elements should be taken into consideration during the planning phase:

- nature, duration, complexity;
- stages;
- verification & validation;
- responsibilities and authorities;
- resource needs;
- interfaces;
- involvement of customers and users;
- requirements for provision of product & services;
- expected controls;
- documented information.

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- Design and development inputs -



The organization is required to determine the inputs for the design and development process.

Inputs = clear, unambiguous, complete and adequate.

Elements to be taken into consideration:

- functional and performance requirements;
- information from previous similar design and development activities;
- statutory and regulatory requirements;
- standards or codes of practices;
- potential consequences of failure.

The organization has to retain documented information on design and development inputs.

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- Design and development controls -



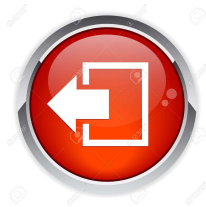
The organization is required to control the design and development process and apply measures to ensure that this process is effective.

The organization has to ensure that:

- the results to be achieved are defined;
- reviews are conducted;
- verification is performed;
- validation activities are conducted;
- actions are taken on problems found;
- documented information is retained.

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- Design and development outputs -



Design and development outputs give information to processes involved in the provision of product and service (production, purchasing, delivery, post-delivery, end of life disposal).

Design and development outputs have to:

- be consistent with the inputs;
- be adequate to the processes of product and service provision;
- Include or reference to monitoring and measuring requirements as well as acceptance criteria;
- Specify the characteristics of the products and services.

The organization is required to retain documented information on the design and development outputs.

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- Design and development changes -



The organization is required to identify, review and control changes made during, or subsequent to, the design and development of products and services, to the extent necessary to ensure that there is no adverse impact on conformity to requirements.

Documented information shall be retained with regards to:

- design and development changes;
- results of reviews of changes;
- authorization of the changes;
- actions taken to prevent adverse impacts.

ISO 9001:2015 Quality Management System Auditor

- Design and development changes -



AUDITING DESIGN AND DEVELOPMENT

Understand exactly the activity of the organization and evaluate whether design and development is applicable in full (with all its requirements) or only part of the requirements.

Is there a planning of design and development?

(Responsibilities for all activities of the design and development – planning, planning, reviewing, verification, validation, approval of changes have been assigned?)

Design and development inputs – have been defined?

Controls for the design and development – in place, what are those controls?

Design and development outputs – defined?

(documented information for all aspects)

Changes to design and development – review, authorization, actions to prevent undesired effects – documented information.

ISO 9001:2015 Quality Management System Auditor

- Control of externally provided processes, products and services -



The organization needs to ensure that externally provided processes, products and services conform to requirements.

External providers:

- suppliers,
- subcontractors,
- associate companies,
- corporate headquarters in case of multi site organizations, etc.

Controls for products, services & processes have to be applied when:

- they are incorporated in the organization's own product & services;
- They are delivered to the customer directly on behalf of the organization;
- A process of part of it is performed by an external provider.

ISO 9001:2015 Quality Management System Auditor

- Control of externally provided processes, products and services -



The organization is required to determine and apply criteria for the evaluation, selection, monitoring of performance and re-evaluation of its external providers.
&
Documented information needs to be retained.

TYPE AND EXTENT OF CONTROL

Type and extent of controls applied to processes, products and services as well as to the external provider depend on their impact over conformity

↑
RISK BASED THINKING

ISO 9001:2015 Quality Management System Auditor

- Control of externally provided processes, products and services -



Examples of controls for processes provided by external providers:

- ensure that the personnel of the external provider has the needed qualifications;
- inspection performed at the site of the external provider by the organization or by a third party contracted by the organization;
- testing of products provided in a laboratory.

Examples of verifications for processes, products and services:

- inspection of products received,
- review of conformity documentation (certificates, test reports);
- second party audits – audits performed by the organization at its suppliers;
- testing – on samples of products or on all products;
- review of statistical data or performance indicators.

ISO 9001:2015 Quality Management System Auditor

- Control of externally provided processes, products and services -



INFORMATION FOR EXTERNAL PROVIDERS

The organization has to ensure that it communicates clearly to external providers the requirements and the controls needed for the processes, products and services it intends to purchase.

Requirements communicated to external providers:

- requirements for the processes, products and services;
- competence required including qualifications of personnel;
- interactions of the external provider with the organization;
- controls applied to the external provider by the organization to monitor its performance;
- verification or validation activities.

ISO 9001:2015 Quality Management System Auditor

- Control of externally provided processes, products and services -



AUDITING

- The policy of the organization for evaluating, selecting, monitoring and re-evaluating suppliers, the criteria used for this process and the documented information that sustain this processes.
- The use of the risk based approach.
- The systems used by the organization to communicate their requirements to external providers.
- The system used by the organization to verify that its requirements for the processes, products and services have been fulfilled.

Review of documented information (*list of suppliers, supplier evaluation, orders, conformity documents, testing reports, communication with suppliers, etc*)

Interviews.

ISO 9001:2015 Quality Management System Auditor

- Control of production and service provision -



Production and service provision have to be performed in controlled conditions.

Controlled conditions:

- available documented information;
- available suitable monitoring and measuring resources;
- available infrastructure and environment for the operation of processes;
- competent persons;
- validation and re-validation;
- actions to prevent human error;
- release, delivery and post-delivery activities.

ISO 9001:2015 Quality Management System Auditor

- Control of production and service provision -



AUDITING

- *Review of documented information* describing the characteristics of the products and services as well as how to perform the activities.
- Measurement and monitoring resources available as well as infrastructure and environment for operation of processes – *Observation*
- Personnel is competent – *Review of documented information*
- Validation required for processes – how is it performed – *documented information review and interviews*
- Actions to prevent human error – *observation + interviews*

ISO 9001:2015 Quality Management System Auditor

- Identification and traceability -



The organization is required to apply suitable means for the identification and traceability so that it is able to determine the processes, products or services that may be affected by nonconformities.

The organization has to define:

- Why the outputs of its processes need to be identified (for example legal requirements in the food industry or in aerospace).
- At which stages of the process the identification is made and how.

ISO 9001:2015 Quality Management System Auditor

- Identification and traceability -



Possible identification methods:

- a code or a title;
- a number of the batch
- a sign
- a system to number electronic documented information

In some cases the ability to go back and trace the outputs is critical.

The organization needs to retain documented information necessary to enable traceability.

AUDITING

What is the system used by the organization to identify the outputs of its processes.

Legal or regulatory requirements for the traceability?

Review of documented information on traceability.

Make an exercise to check traceability for a product.

ISO 9001:2015 Quality Management System Auditor

- Property belonging to customers or external providers -



The organization is required to exercise care with property belonging to customers or external providers while it is under its control or being used by it.

Property belonging to customers and external providers needs to be:

- identified,
- protected and
- safeguarded.

Property of customers or external providers



Tangible (products, components, materials, equipment, premises) or intangible (like personal information) and the actions to protect it depend on its characteristics

ISO 9001:2015 Quality Management System Auditor

- Property belonging to customers or external providers -



Examples of measures to protect property of customers or external providers:

- Assign owner
- Store it in a designated place
- Control access
- Delete at the end of project, etc

Risk based thinking should be applied – to take into consideration the criticality of the property belonging to customers and external providers.

In case the property of the customer or external provider is lost, damaged or otherwise found unsuitable for use the organization needs to report this to the customer or external provider and retain documented information on the matter.

ISO 9001:2015 Quality Management System Auditor

- Property belonging to customers or external providers -



AUDITING

The organization uses property that belongs to its customers or external providers? What is it?

What controls are in place to identify it, protect it and safeguard it?

Cases when the property has been affected?

Documented information should be available.

ISO 9001:2015 Quality Management System Auditor

- Preservation -



The organization shall preserve the outputs to the extent necessary to ensure conformity to requirements.

Preservation can involve – identification, handling, contamination control, storage, transmission, transportation and protection.

Manufacturing:

- Storing the products in warehouses
- Identifying products
- Ensure integrity & security
- Transport & handle.

Services:

- Ensure specific conditions
- Provide specific environment

ISO 9001:2015 Quality Management System Auditor - Preservation -



AUDITING

Observe preservation conditions and methods

Legal or regulatory requirements applicable to the preservation in the industry.

Monitoring of preservation conditions using equipment

Preservation conditions applicable to the customer or end-user – are they communicated?

ISO 9001:2015 Quality Management System Auditor

- Post-delivery activities -



Delivery may not end organization's responsibility for the products and services.

To determine post-delivery activities required the following are to be taken into consideration:

- statutory and regulatory requirements;
- customer requirements;
- customer feedback;
- potential undesired consequences associated with its products and services;
- the nature, the use and the intended lifetime of its products and services.

ISO 9001:2015 Quality Management System Auditor

- Post-delivery activities -



Examples of post delivery may include:

- Communication with the customer to determine if the product and services were to their satisfaction;
- Installation of an equipment bought by the customer, training of the operators and disposal of the user's old equipment;
- Warranty or technical support;
- Maintenance or other contractual obligations;
- Recycling of the product at the end of its lifetime or final disposal

ISO 9001:2015 Quality Management System Auditor - Post-delivery activities -



AUDITING

Are there post-delivery activities?

Are they performed as agreed with the customer?

Review of documented information (*warranty reports, installation reports, training minutes, maintenance records, etc*)

When post-delivery activities are outsourced the auditor will need to evaluate the methods used by the organization to monitor the performance of the external provider

ISO 9001:2015 Quality Management System Auditor

- Control of changes -



Changes to the production and service provision processes need to be reviewed and controlled to ensure that the outputs, products and services continue to conform with requirements.

Risk based thinking should be applied when deciding to change and the potential effects of change have to be evaluated by the organization.

Changes need to be examined before implementation.

Documented information shall be retained on:

- results of the review of change;
- person(s) authorizing the change;
- action(s) generated from the review of change.

ISO 9001:2015 Quality Management System Auditor

- Control of changes -



AUDITING

Change management process: responsibilities, authorities, risk evaluation.

Review of documented information

(ex. review minutes, risk analysis, verification and validation results, description of the changes to be implemented, best case and worst case scenarios).

ISO 9001:2015 Quality Management System Auditor

- Release of products and services -



The organization is required to ensure that products and services conform to all requirements applicable before being released to the customer.

The organization shall retain documented information on the release of products and services :

- evidence of conformity with criteria for the products or services;
- the person or persons authorizing the release.

The person or persons authorizing the release need to be nominated and clearly identified.

ISO 9001:2015 Quality Management System Auditor

- Release of products and services -



AUDITING

Review of documented information that demonstrates the products and services do conform to all applicable requirements

(e.g. testing reports, checklists, etc depending on the type of the product).

Person or persons who authorized the release can be traced back using the documented information available?

Review of documented information accompanying the product or service released *(e.g. performance declaration, conformity declaration, etc)*

ISO 9001:2015 Quality Management System Auditor

- Control of nonconforming outputs -

A red tag with a white string loop at the top left. The tag has the text 'NON-CONFORMING' in bold black letters at the top. Below this, there are several lines of text with blank spaces for information: 'Part No. _____ Serial No. _____', 'Part Name _____', 'Customer _____', 'P.O. No./Date _____ No. of Pieces _____', 'Reason _____', and 'Inspector _____ Date _____'.

The organization is required to ensure that outputs that do not conform to their requirements are identified and controlled to prevent their unintended use or delivery.

When a nonconforming output is determined the organization should take appropriate action based on the effect on the conformity of the product or service

Dealing with nonconforming outputs:

- correction;
- segregation, containment, return or suspension of provision of products and services;
- inform the customer;
- obtain authorization for acceptance under concession.

ISO 9001:2015 Quality Management System Auditor

- Control of nonconforming outputs -



In cases where nonconforming products and services are corrected they have to be verified to ensure that it is conform to requirements.

Documented information needs to be retained that:

- Describes the nonconformity;

- Describes the actions taken to correct, mitigate, communicate;

- Describes any concessions obtained and

- Identifies the authority deciding the action in respect to the nonconformity.

Documented information - *nonconformity reports or other such forms; databases of nonconforming outputs; mobile apps; etc*

ISO 9001:2015 Quality Management System Auditor

- Control of nonconforming outputs -



AUDITING

- What is considered by the organization to be nonconforming output?
- What are the actions to be taken in case such an output is identified?
- Who is responsible for recording, reviewing and deciding with regards to a nonconforming output.
- Documented procedures?

Risk based thinking aimed at the criticality of nonconforming outputs?

- Review of documented information— describing the nonconformities, the actions taken, the person or persons having the authority on deciding on the nonconformity as well as any concessions obtained if the case.

Interview – to evaluate how nonconforming products are identified and managed.

- Also nonconforming outputs should be evaluated by the organization in terms of trends and used as a method for improvement.

ISO 9001:2015 Quality Management System Auditor

- Monitoring, measurement, analysis and evaluation -



Monitoring, measurement, analysis and evaluation - to determine if the results expected are being achieved.

Determine:

- What needs to be monitored and measured;
- What methods are used
- When to perform monitoring and measuring and
- When the results are being analyzed and evaluated.

ISO 9001:2015 Quality Management System Auditor

- Monitoring, measurement, analysis and evaluation -



Evaluate the performance and the effectiveness of the QMS

Performance = the measurable results of the organization.

Effectiveness = extent to which planned activities are realized and planned results are achieved.

The organization is required to retain documented information as evidence of the results of monitoring, measurement, analysis and evaluation.

ISO 9001:2015 Quality Management System Auditor

- Monitoring, measurement, analysis and evaluation -



AUDITING

What the organization considers needed to monitor and measure.

What methods are used.

Examples: key performance indicators (for the processes), conformity of the products and services, accomplishment of quality objectives, customer satisfaction, warranty claims, complaints, etc.

Evaluate if the results of monitoring and measurement are analyzed and evaluated.

(management review)

Review of documented information – results of monitoring, measurement, analysis and evaluation.

RiG
CERT
TRAINING

[illegible]

Customer satisfaction info = input to management review

ISO 9001:2015 Quality Management System Auditor - Customer satisfaction -



AUDITING

What are the methods used by the organization to collect information on customer satisfaction?

How is the information used?

What actions are generated?

Customer satisfaction info may be a source of opportunities & risks.

ISO 9001:2015 Quality Management System Auditor

- Analysis and evaluation -



Data and information collected through monitoring and measuring has to be analyzed and evaluated to determine if processes, products and services meet requirements and identify actions and opportunities for improvement.

Data used for analysis is up to the organization and it can be:

- Data related to the products
- Data related to service performance
- Results from the monitoring of customer satisfaction
- Data regarding the performance of the QMS
- If the planning has been implemented
- The effectiveness of actions to treat risks and opportunities
- The performance of external providers
- The status of quality objectives
- The need to improve the QMS

ISO 9001:2015 Quality Management System Auditor

- Analysis and evaluation -



Frequency = up to the organization.
Statistical techniques can be used.

Data generated from analysis and evaluation can be in different forms *trend analyses, balance scorecards, dashboards – and it should be input to management review.*

AUDITING

Methodology used by the organization to collect the data and information that is used for analysis and evaluation.

What information and data is taken into consideration?

Statistical techniques?

Results are provided in a form that is useful to the organization?

Information is used in management review and other processes?

ISO 9001:2015 Quality Management System Auditor

- Internal audit -



The organization is required to perform internal audits of its QMS at planned intervals.

Audit programme covers the internal audits for a given period of time.

Frequency of internal audits = Risk based thinking
and

- importance of the processes
- priorities of the management
- changes that affected the organization
- legal compliance issues
- results of past audits (internal or external)
- trends in nonconforming outputs or customer complaints.

ISO 9001:2015 Quality Management System Auditor

- Internal audit -



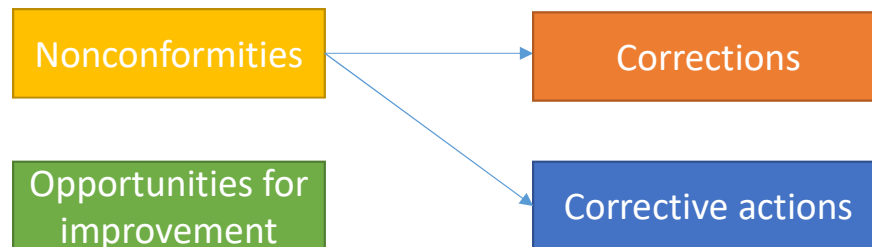
Auditors = objectivity + impartiality

Recommended – audit requirements of ISO 9001 applied to a process/ project.

Results of internal audits (audit report)

- reported to management;
- input to management review

ISO 19011 – guidelines
for internal audits



The organization is required to retain documented information as evidence of internal auditing.

ISO 9001:2015 Quality Management System Auditor

- Internal audit -



AUDITING

- The internal audit programme or programmes + risk based approach
- The competence of internal auditors + impartiality & objectivity
- Documented information generated by the internal audit process – plans, checklists, reports, nonconformity reports
- Actions taken on the nonconformities and the implementation of corrections and corrective actions.
- If the results of internal audits are used as input data into the management review.

ISO 9001:2015 Quality Management System Auditor

- Management review -



Top management is required the QMS at planned interval.
Why? To ensure it continues to be:
continues to be:

- suitable;
- adequate;
- effective and
- in alignment with the strategic direction of the organization.

Management review meetings should take place at planned intervals.

Standalone or along with other activities.

ISO 9001:2015 Quality Management System Auditor

- Management review -



Management review inputs:

- status of actions from previous management reviews
- changes in the external and internal issues
- information on the performance and effectiveness of the QMS including trends in:
 - customer satisfaction and feedback from relevant interested parties;
 - the extent to which quality objectives have been met;
 - process performance and conformity of products and services;
 - nonconformities and corrective actions;
 - monitoring and measurement results;
 - audit results;
 - the performance of external providers;
- the adequacy of resources;
- the effectiveness of actions taken to address risks and opportunities;
- opportunities for improvement.

ISO 9001:2015 Quality Management System Auditor

- Management review -



Management review outputs = decisions and actions should be related to:

- opportunities for improvement;
- changes needed for the QMS and
- resource needs.

The organization is required to retain documented information as evidence of the results of management review.
(minutes of the meetings, reports, plans of actions)

ISO 9001:2015 Quality Management System Auditor

- Management review -



AUDITING

Management review meetings do cover at least all the aspects specified as inputs and outputs?

Actions and decisions generated from the management review are aimed at the improvement of the QMS?

Review of documented information as well as interviews with top management are methods that can be used by the auditors for the evaluation of this requirement

ISO 9001:2015 Quality Management System Auditor

- Improvement -



The organization is required to determine opportunities for improvement, plan and implement the necessary actions in order to achieve intended results as well as enhance customer satisfaction.

Actions should look to:

- improve the products and services;
- correct, prevent or reduce undesired effects and
- improve the performance and effectiveness of the QMS.

ISO 9001:2015 Quality Management System Auditor - Improvement -



Improvement can be conducted in different methods like:

- implementing measures to prevent the recurrence of nonconformities;
- incremental improvement to processes or products and services;
- implementing major changes to existing processes or new processes;
- use new technologies or innovations.

AUDITING

The organization has identified opportunities for improvement?

Any actions have been taken on those opportunities?

(actions to address risks and opportunities)

Possible sources of information: internal audits, management reviews; measuring and monitoring (for example of customer satisfaction), complaints, trends in nonconforming outputs.

ISO 9001:2015 Quality Management System Auditor

- Nonconformity and corrective action -



The organization is required to react to nonconformities and implement corrective actions whenever nonconformities are identified.

React to the nonconformity by taking actions to control and correct it and deal with the consequences.

Evaluate the need for action to eliminate the cause of the nonconformity so that it does not recur or occur somewhere else

Implement the action needed

Review the effectiveness of the corrective actions taken

Update risks and opportunities if necessary

Make changes to the QMS, if necessary

Retain documented information on:

- nature of nonconformities and actions taken
- results of corrective actions

ISO 9001:2015 Quality Management System Auditor

- Nonconformity and corrective action -



AUDITING

Review documented information about the nonconformity management process as well as conduct interviews to see if personnel is aware of the methods used to identify and treat nonconformities.

Documented information may include:

nonconformity reports detailing the nonconformities found, their severity, their root cause analysis, the actions taken to deal with the consequences, the corrective actions implemented and the verification of their effectiveness.

Investigation of the nonconformity being present in other processes or in other areas of the organization.

Risks and opportunities and actions to address them have been revised following the occurrence of the nonconformity.

ISO 9001:2015 Quality Management System Auditor

- Continual improvement -



The organization shall continually improve the suitability, adequacy and effectiveness of the QMS.

Sources to identify opportunities for improvement:

- the results of analysis and evaluation;
- management review; etc.



Organization should look to:

- increase the level of conforming outputs,
- improve its processes and reduce process variation in order to enhance general performance for the benefit of its customers and interested parties.

ISO 9001:2015 Quality Management System Auditor

- Management system auditing -



Quality management system is:

a set of processes, policies and procedures needed by an organization to plan and perform its activities (regardless whether its production or service provision).

Management system audit

- Systematic,
- Independent and
- Documented



process to obtain audit evidence
and

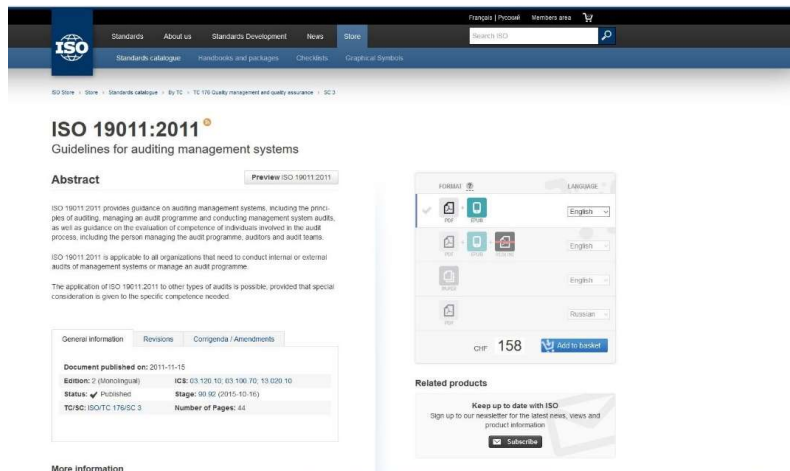
evaluating it objectively to determine the extent to which
audit criteria are fulfilled.

ISO 9001:2015 Quality Management System Auditor - Management system auditing -



ISO 19011 - Guidelines for auditing management systems

Standard dedicated to management system auditing



www.iso.org

ISO 9001:2015 Quality Management System Auditor

- Management system auditing -

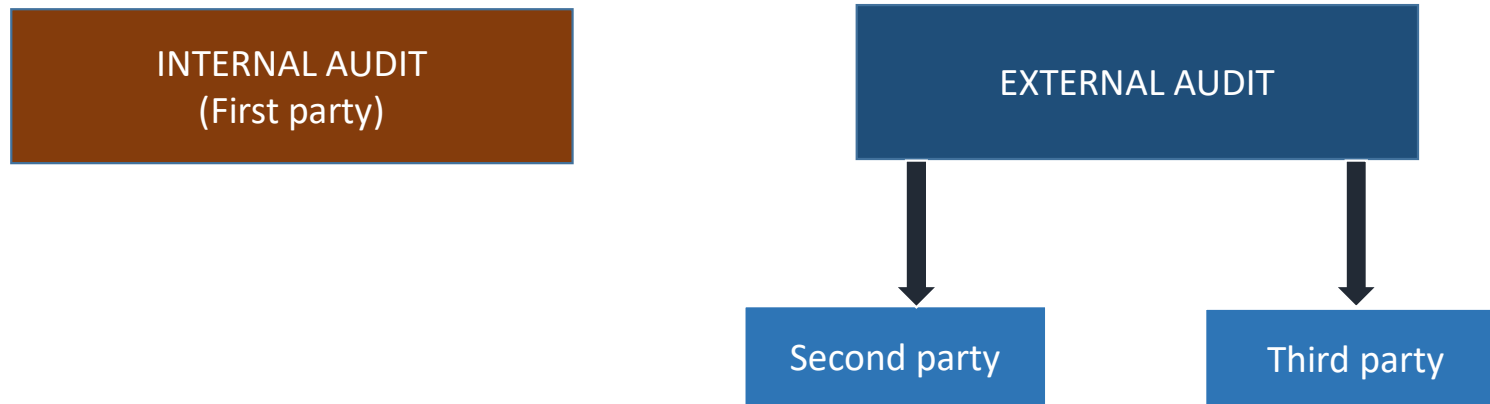


Principles of management system auditing:



ISO 9001:2015 Quality Management System Auditor

- Types of management system audits -



Audit client

Who requests the audit

Auditee

Organization being audited

ISO 9001:2015 Quality Management System Auditor

- Types of management system audits -



INTERNAL AUDIT (First party)

Done by the organization itself – using own auditors or contracted

EXTERNAL AUDIT (Second party)

Audit conducted by parties that have an interest in the organization (e.g. customers on their suppliers)

EXTERNAL AUDIT (Third party)

Audit conducted by independent organizations - for certification, for legal or regulatory reasons.

ISO 9001:2015 Quality Management System Auditor

- Audit scope, objectives and criteria -



Audit objectives – what is to be accomplished by the audit

Examples:

- determine the conformity of the management system with audit criteria;
- evaluate the capability of a management system to ensure compliance with legislation
- identification of areas for improvement

Audit scope – where, when and what

Usually includes - locations where the audit will take place, time periods when the audit is performed and what activities/ processes are audited

Audit criteria – reference against which conformity is determined

- Standard ISO 9001:2015
- Internal procedures
- Contract obligations
- Legal requirements/ other requirements

ISO 9001:2015 Quality Management System Auditor

- Audit team -



Audit team is selected taking into consideration competence needed for achieving audit objectives.

Audit team

Can comprise one or more persons.

Should have at least one auditor that acts as lead auditor and coordinates the activity of all members

May also include: auditors, auditors in training, technical experts, guides, observers

ISO 9001:2015 Quality Management System Auditor

- Audit plan and working documents -



Audit plan

Each audit shall have an audit plan.

Is done by under the authority of the lead auditor and should includes at least:

- objectives and criteria
- composition of the audit team,
- locations,
- processes,
- dates and expected duration of audit,
- methods used during the audit,
- roles of the audit team members,
- representatives of the audited organization,
- aspects about the audit report,
- logistic and communication aspects (transport of audit team if the case),
- language (if this topic is relevant).

Audit plan is agreed with the auditee
prior to commencing the audit.

Working documents

- checklists and any other documents to record evidence;
- forms to record participation
- forms to record nonconformities.

Working documents are prepared before the audit and distributed to the members of the audit team

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- Audit activities -



Opening meeting



Audit starts with the opening meeting.

It is performed to:

- confirm the agreement of all parties over the audit plan
- present the members of the audit team
- ensure that all activities planned can be performed

Participants are:

audit team members and
representatives of the audited organization

ISO 9001:2015 Quality Management System Auditor

- Audit activities -



Audit methods

- Observation of activities;
- Interview
- Review of documented information.

Auditors shall record evidence they obtain – whether to sustain conformity or nonconformity.

Findings

Findings of the audit are the results of evaluating audit evidence collected with audit criteria

They refer to the conformity or nonconformity with every requirement of the audit criteria

(the standard for example – ISO 9001)

Conclusions

The result of the audit after taking into consideration the objectives of the audit and the findings

ISO 9001:2015 Quality Management System Auditor

- Audit activities -



Closing meeting

At the end of the audit to present findings and conclusions.
Conducted by the lead auditor

Closing meeting will discuss:

- that the information was collected by the audit team using sampling;
- that an audit report will be elaborated and communicated to the organization, and possibly other parties (audit client);
- findings and conclusions of the audit so they are understood by the auditee
- post-audit activities – like follow up audits to evaluate corrections and corrective actions.

Participants are:

audit team members and
representatives of the audited organization.

ISO 9001:2015 Quality Management System Auditor

- Nonconformities -



Nonconformities

Nonfulfillment of a requirement of the audit criteria.

Nonconformities should be formulated clear enough, sustained with evidence and related to a requirement of audit criteria.

The goal of management system audit is NOT to find nonconformities but to help IMPROVE

May be **MAJOR and MINOR**.

MAJOR – affect the functioning of the whole management system

MINOR – do not affect the functioning of the system

Opportunities for improvement may be recorded if agreed so.

ISO 9001:2015 Quality Management System Auditor

- Audit report -



Audit report is distributed to: auditee, audit client (if they are different), certification body, etc.

Audit is completed when all planned activities have been finalized.

Audit report

Made by the lead auditor.

Includes:

- objectives,
- scope,
- criteria,
- dates and locations where the audit was conducted,
- audit findings and evidence,
- audit conclusions,
- identification of the auditee and audit client,
- follow-up plans,
- confidentiality aspects.